Suppose we allow an interaction between the treatment and the conditioning variable

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$

In this model, β_1 can *no longer* be interpreted predictively as the difference between two conditional expectations. Instead, we predictively interpret $(\beta_1 + \beta_3 \cdot x)$ as the difference between two conditional expectations $E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]$ which depends on the value of $X = x$. This can be most easily seen by rearranging terms in the conditional expectation:

$$E[Y|T = t, X = x] = \beta_0 + \boldsymbol{(\beta_1 + \beta_3 \cdot x)} \cdot t + \beta_2 \cdot x$$

Conditional exchangeability allows us to interpret $(\beta_1 + \beta_3 \cdot x)$ causally (in terms of potential outcomes):

$$\begin{aligned}
(\beta_1 + \beta_3 \cdot x) &= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t, x \\
&= E[Y|T = t + 1, X = x] - E[Y|T = t, X = x] \text{ for all } t \\
&= E[y^{t+1}|T = t + 1, X = x] - E[y^t|T = t, X = x] \text{ for all } t \\
&= E[y^{t+1}|X = x] - E[y^t|X = x] \text{ for all } t \\
&= E[y^{t+1} - y^t|X = x] \quad \text{for all } t \qquad (1)
\end{aligned}$$

However, although the average treatment effect conditional on $X = x$ ($E[y^{t+1} - y^t|X = x]$) may be of substantive interest, the law of total expectation, and our linearity assumption allow us to interpret $(\beta_1 + \beta_3 \cdot E[X])$ as the average treatment effect.

$$\begin{aligned}
\beta_1 + \beta_3 \cdot E_X\{X\} &= E_X\{\beta_1 + \beta_3 \cdot X\} \\
&= E_X\{E[Y|T = t + 1, X = x] - E[Y|T = t, X = x]\} \\
&= E_X\left\{E[y^{t+1}|T = t + 1, X = x] - E[y^t|T = t, X = x]\right\} \\
&= E_X\left\{E[y^{t+1}|X = x] - E[y^t|X = x]\right\} \\
&= E_X\left\{E[y^{t+1} - y^t|X = x]\right\} \\
&= E[y^{t+1} - y^t]
\end{aligned}$$

The variance is more difficult. We know that conditional on $X = x$, we have a marginally randomized experiment such that the standard regression estimate of variance for $(\hat{\beta}_1 + \hat{\beta}_3 \cdot x)$ will tend to be conservative. However, the variance for the